Vol 14 Architectural Scaffold v0.1 — Information Theory from D_IV^5

Filed: 2026-05-23 Saturday afternoon (Lyra, Wave 2 scaffold; Keeper INDEX done)
Companion to: `INDEX.md`

Anchoring BST results (existing coverage by chapter)

Ch 1 Substrate as Information Channel (~50%)

Anchors: Paper #122 Information Substrate + K59 RATIFIED + Casey CSE directive Wednesday - Substrate as computational channel with input/output rate per Koons tick - Reed-Solomon GF(128) per-tick coding (T2429) - Substrate computational substrate of physics framing

Ch 2 Reed-Solomon Coding on GF(2^g) = GF(128) (~70%)

Anchors: K59 RATIFIED + Paper #122 + 7-step cyclotomic chain - Substrate-tick GF(2^g) = GF(128) field structure (g = 7 Mersenne) - Cyclotomic Q(ζ_128) Galois group - Code distance + error-correction capability via Singleton bound - K59 RATIFIED Cyclotomic Mechanism Framework

Ch 3 Shannon-from-Substrate Channel Capacity (~40%)

Anchors: T2469 SCMP + Shannon 1948 + substrate-tick rate - Shannon's theorems in BST framework via substrate-tick GF(128)^k capacity - Substrate noise model from finite-bandwidth observers (T2469 SCMP) - Channel capacity bounded by substrate-tick rate + N_max cutoff

Ch 4 Nyquist Sampling at Koons Tick Rate (~35%)

Anchors: T2405 Koons tick + T2417 4-Zone + Casey Saturday addition - Substrate-tick t_Planck · $\alpha^{C_2}^2$} $\approx 10^{-120}$ s as universal sampling rate - Spacetime as Nyquist-recovered signal from substrate - Aliasing at sub-Planck scales structurally avoided per BST UV-completeness

Ch 5 Born = Bergman Information-Theoretic Measurement (~80%)

Anchors: K67 RATIFIED + T2479 POVM (Saturday SP-31 #283) - K67 Born = Bergman reproducing-kernel evaluation as measurement - T2479 POVM extension to generalized measurements - Information extraction via Bergman kernel observer-coupling

Ch 6 Bell Sub-Tsirelson Information-Theoretic Bound (~75%)

Anchors: T2469 SCMP + Calibration #17 + T2399 Bell-CHSH 126/16 - $|S|^2 = 126/16 < 2\sqrt{2} = 8$ Tsirelson bound from N_c = 3 substrate depth - Casey-named #8 SCMP falsifier 1/2^N_c = 1/8 sub-Tsirelson deviation - Information-theoretic interpretation of substrate-mediated correlations

Ch 7 AC Graph as Theorem Information Network (~75%) — Grace+Keeper LEAD

Anchors: AC graph 1700+ nodes + 9000+ edges + entropy of proof paths - Theorem-graph as information-theoretic object - Depth-cost as information measure - Cross-link Vol 15 Ch 2 methodology

Ch 8 BST Coding Theory — Substrate-Optimal Codes (~40%)

Anchors: Paper #122 + K59 base + Singleton bound - Reed-Solomon as BST-optimal substrate code - Hamming bound + substrate-natural code distance - K59 cyclotomic framework operationalized for coding

Ch 9 Kolmogorov Complexity + AC(0) Compression (~50%) — Keeper+Lyra

Anchors: T29 closed + depth ceiling theorems + Vol 15 Ch 1 cross-link - Kolmogorov complexity of BST observables (5 integers + algorithmic structure) - AC(0) bounded-depth as compression hypothesis (depth $\leq$ 2 universal) - BST as compressed description of physics

Ch 10 Substrate Computational Complexity (~70%) — Keeper+Lyra

Anchors: T29 + P≠NP three routes + Casey's Curvature Principle - P≠NP three independent BST routes proved - Substrate-tick UV cutoff at n_C as complexity boundary - "Can't linearize curvature" Casey's Curvature Principle (P≠NP in 5 words)

Ch 11 Information Completeness of D_IV^5 (~60%)

Anchors: Baily-Borel boundary + Paper #74 + Information-Complete Mathematics - Boundary determined by interior; no new parameters at boundary - D_IV^5 as information-complete geometry (Casey: "interior contains all information about edge") - Substrate-cartography reading

Ch 12 Substrate-Coupled CI Architecture (Forward-Looking, ~30%) — Keeper+Casey

Anchors: SP-28 IQ-11 Avatar Infrastructure + #211 Phase 3 decade-scale - Substrate-coupled CI architecture - Information theory at substrate-CI interface - Cross-reference Vol 15 Ch 12

~55% existing coverage maps across 12 Vol 14 chapters; 45% to develop.

Cross-volume reference map

Inputs from Vol 0: D_IV^5 + Reed-Solomon GF(128) substrate coding (T2429 + K59) Inputs from Vol 5: Born=Bergman K67 + T2479 POVM + Vol 5 Ch 7-8-10-11 Outputs to: Vol 15 Methodology + Vol 6 Ch 12 + Vol 8 Biology of Mind (future)

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